Mathematical Foundations of Political Methodology

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- Integral
- Fundamental Theorem of Calculus
- Integration Facts
- Indefinite and Definite Integrals

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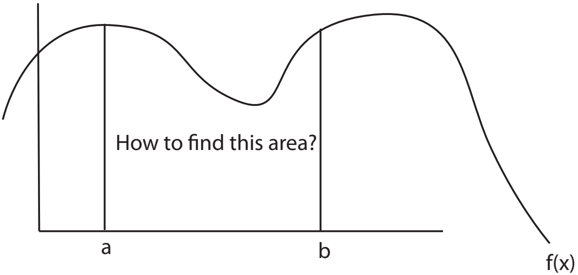
Integral: Geometric Intuitions

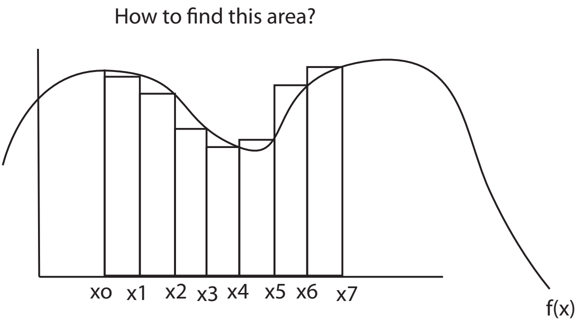
"The opposite of differentiation," "antiderivative," or "taking an area"



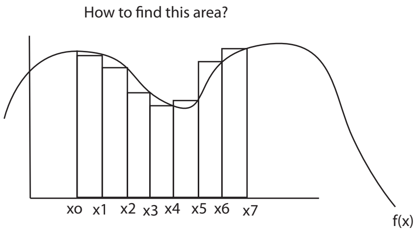
Integral: Geometric Intuitions

Let $a < b$ be real numbers and let f be a continuous function.

We want to find the area bounded by the curve $y = f(x)$, the vertical lines $x = a$ and $x = b$, and the x-axis. Let $a = x_o$ and $b = x_7$



Integral: Geometric Intuitions



$\sum_{i=1}^{7}(x_i - x_{i-1})f(x_i)$ or letting $\Delta x_i = x_i - x_{i-1}$ this approximation is $\sum_{i=1}^{7} f(x_i)\Delta x_i$

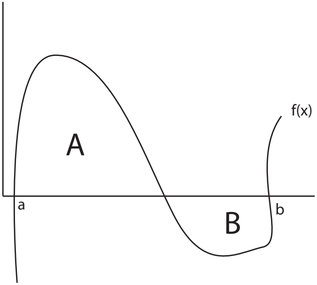
Letting $\lim_{\Delta x_i} \to 0$ gives us the solution; it's written

$$\int_a^b f(x)dx$$

Integral: Geometric Intuitions

Areas below the $x-$axis are negative:

$$\int_a^b f(x) = \text{Area(A)-Area(B)}$$



Integral: Definition

Definition

Suppose $f : \mathbb{R} \to \mathbb{R}$. We will define the Riemann Integral as $\int_a^b f(x)dx$. If this exists then we say f is integrable on $[a, b]$ and call $\int_a^b f(x)dx$ the integral of f.

Features of an Integral

$$\int_a^b f(x)dx$$

- $\int$ is the indication for taking an integral.
- dx tells us that we are taking an integral with respect to x.
- a is the lower bound of the integration.
- b is the upper bound of the integration.

Example of Non-Integrable Function

Suppose $f : [0,1] \to \frac{1}{x}$

$$\int_0^1 \frac{1}{x} dx$$

Then $\frac{1}{x}$ is not integrable on $[a,b]$ because the area that the integral would represent is infinite.

$$f(x) = \begin{cases} 1 \text{ if } x \text{ rational} \\ 0 \text{ if } x \text{ irrational} \end{cases}$$

This function is called Dirichlet Function, it is not integrable, because every interval will contain a discontinuous jump

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Fundamental Theorem of Calculus

There is a connection between derivative and integral, the relationship is descried by the Fundamental Theorem of Calculus. First, we introduce the antiderivative F:

Theorem

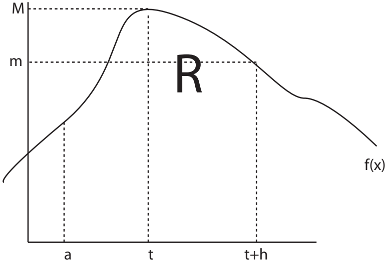
Given a number a and a continuous function $f(x)$ we can define another function F such that

$$F(t) = \int_a^t f(x)dx \text{ for all } t$$

Then F is differentiable and $F'(t) = f(t)$ for all t

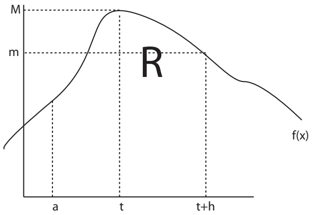
If you integrate a function and then differentiate the result, you retrieve the function that you started with $\Rightarrow$ "integration is the reverse of differentiation"

The Geometry of the Antiderivative F



- $f(t+h) = m$, $f(t) = M$
- We will use mean value theorem to show that between t and $t+h$, there must be a box equal to the area under the curve.

The Geometry of the Antiderivative F



Let $R = \int_t^{t+h} f(x)dx$ and let $F(t) = \int_a^t f(x)dx$ for all t. Then

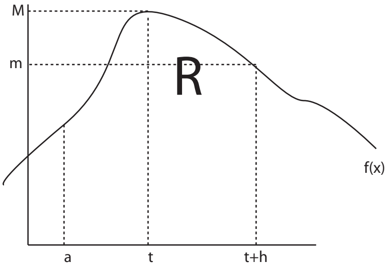
$$R = \int_a^{t+h} f(x)dx - \int_a^t f(x)dx = F(t+h) - F(t)$$

The box made by h*m is smaller than R, the box made by h*M is bigger.

$$hm \leq F(t+h) - F(t) \leq hM$$

$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

The Geometry of the Antiderivative F



As $h \to 0$ we have $m \to f(t) = M$. Thus,

$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

implies that

$$\lim_{h \to 0} \frac{F(t+h) - F(t)}{h} = f(t)$$

or that

$$F'(t) = f(t) \quad \square$$

Fundamental Theorem of Calculus

Theorem

Proposition: If f has a continuous derivative, then

$$\int_a^b f'(x) = F(b) - F(a)$$

$f(x)$ is the derivative of F(x).
$F(x)$ is the antiderivative of $f(x)$.
Key idea, antiderivatives are not unique.

Fundamental Theorem of Calculus

Start with $F' = f$ Define: $g(x) = \int_a^x f(t)dt$.
By the first bit, $g(x)$ is also an antiderivative of f on $[a, b]$.
$f(x) = g'(x) = F'(x) \rightarrow g'(x) - F'(x) = 0$
So $(g(x) - F(x))' = 0$
Recall, from the mean value theorem for $c < t < d$:

$$\frac{f(d) - f(c)}{d - c} = f'(t)$$

$$f(d) = f(c) + (d - c)f'(t)$$

if f'(t)=0, then

$$f(d) = f(c)$$

f() is constant!.

Fundamental Theorem of Calculus

$$(g(x) - F(x))' = 0 \to g(x) - F(x) \text{ is constant.}$$

$$g(x) - F(x) = C$$

$$g(b) - F(b) = g(a) - F(a)$$

$$g(b) = g(a) - F(a) + F(b)$$

$$\int_a^b f(t)dt = \int_a^a f(t)dt - F(a) + F(b)$$

$$\int_a^b f(t)dt = 0 - F(a) + F(b)\square$$

Apply Fundamental Theorem of Calculus

$$\int_a^b f^{'}(x)dx \quad = \quad f(b) - f(a)$$

- Calculate antiderivative
- Evaluate at b
- Evaluate at a

Some Classic Antiderivative Formulas

antiderivative = indefinite integral

$$
\begin{aligned}
\int 1 dx &= x + c \\
\int k dx &= kx + c \\
\int x^n dx &= \frac{x^{n+1}}{n+1} + c \\
\int \frac{1}{x} dx &= \log x + c \\
\int e^x dx &= e^x + c \\
\int a^x dx &= \frac{a^x}{\log a} + c
\end{aligned}
$$

Example 1: Uniform Distribution

Suppose $f : R \to R$, with

$$\begin{aligned} f(x) &= 1 \text{ if } x \in [0, 1] \\ f(x) &= 0 \text{ otherwise} \end{aligned}$$

What is the area under $f(x)$ from $[0, 1/2]$?

$$\begin{aligned} \int_0^{1/2} f(x)dx &= \int_0^{1/2} 1dx \\ &= x|_0^{1/2} \\ &= (1/2) - (0) \\ &= 1/2 \end{aligned}$$

We will call $f(x) = 1$ the uniform distribution.

Example 2: Area Under a Line

Suppose $f : R \to R$, with

$$f(x) = x$$

Evaluate the $\int_2^t f(x)dx$

$$\begin{aligned} \int_2^t f(x)dx &= \int_2^t xdx \\ &= \frac{x^2}{2}|_2^t \\ &= \frac{t^2}{2} - \frac{2^2}{2} \\ &= \frac{t^2}{2} - \frac{4}{2} = \frac{t^2}{2} - 2 \end{aligned}$$

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Integration Facts

Theorem

If $f_1, f_2 : [a, b] \to \mathbb{R}$ *and* f_1, f_2 *are integrable on [a,b], then*

i) *Consider the interval* $[a, b]$ *and* $c \in [a, b]$. *Then,*

$$\begin{aligned}
\int_c^c f_1^{'}(x)dx &= f_1(c) - f_1(c) = 0 \\
\int_a^b f_1^{'}(x)dx &= \int_a^c f_1^{'}(x)dx + \int_c^b f_1^{'}(x)dx \\
&= (f_1(c) - f_1(a) + (f_1(b) - f_1(c)) \\
&= f_1(b) - f_1(a)
\end{aligned}$$

Integration Facts

Theorem

If $f_1^{'}, f_2^{'} : [a, b] \rightarrow \mathbb{R}$ and $f_1^{'}, f_2^{'}$ are integrable on [a,b] and $f_1^{'}$ has antiderivative is f_1 and $f_2^{'}$ has antiderivative f_2, then

ii) *For $c_1, c_2 \in \mathbb{R}$ then*

$$\int_a^b (c_1 f_1^{'}(x) + c_2 f_2^{'}(x))dx \quad = \quad c_1 \int_a^b f_1^{'}(x)dx + c_2 \int_a^b f_2^{'}(x)dx$$

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Definite Integrals

With definite integrals we do not need to worry about the constant of integration: We know that

$$\int_a^b g'(x)dx = g(b) - g(a)$$

If we used $g(x) + C$ as our primitive instead we'd get

$$\int_a^b g'(x)dx = (g(b) + C) - (g(a) + C) = g(b) - g(a)$$